DO NOW — RECALL

Lesson 065 · Computational Methods: Recognising Solva

Name: _____ Class: _____ Date: _____

1. What is a precondition?

2. What is abstraction?

3. How does a queue differ from a stack?

4. Trace binary search for 11 in the sorted, zero-indexed array [2,4,6,8,10,12,14]. Start low=0, high=6. Use mid=(low+high) DIV 2 and exclude the checked midpoint when updating bounds. State the checked indexes and the final result.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Condition required for an algorithm to work correctly.

2. Model answer

Selecting relevant details and omitting irrelevant ones.

3. Model answer

FIFO versus LIFO.

4. Model answer

3, 5, 4; not found. Values are 8, 12, 10. The final bounds are low=5, high=4, so the search stops.

By the end, you can…

• A solvable computational problem needs an adequately precise input representation, a finite method and a verifiable output or objective.

• Problem recognition identifies the class of task—classification, searching, route finding, scheduling or optimisation—before selecting a method.

• Define measurable success criteria and constraints; vague goals such as 'best route' require clarification of 'best' (distance, cost or time).

Vocabulary — know these terms

problem recognition — Identify the underlying task type.

constraint — Requirement a solution must obey.

objective function — Quantity to minimise or maximise.

computable specification — Precise task that a program can implement/check.

NEW LEARNING

Read, annotate and explain

Identify

A solvable computational problem needs an adequately precise input representation, a finite method and a verifiable output or objective.

Model

Problem recognition identifies the class of task—classification, searching, route finding, scheduling or optimisation—before selecting a method.

Evaluate

Define measurable success criteria and constraints; vague goals such as 'best route' require clarification of 'best' (distance, cost or time).

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Turn 'choose the best bus route' into a precise computational problem.

Explain the decision

Cost is defined and connections must be valid; output stops and total fare.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define computational problem input, process, output for cheapest route on a network. [6]

Question 2

A pupil asks to 'sort books by best'. Give two precise orderings, explain why different and select one for finding a title. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate whether a computer can choose a 'good teacher' simply from exam results, explaining models, constraints and limits. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Every vague human objective has exactly one obvious numerical optimisation target.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.